

Update: Atom-Photon Gate in QuTiP + Reflection w. Pellical

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Master Thesis Progress Report

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- Goal: simulate an atom–photon CNOT gate using a birefringent cavity with qutip
- Previous limitations:
 - Direct driving of the intracavity field
 - Inconsistent treatment of photon number statistics
 - Fidelity estimates not directly connected to realistic input pulses
- Recent progress: fully input–output based simulation using QuTiP

Input–Output Formalism

- Cavity is driven by an incoming wavepacket (Gardiner–Collett formalism)
- Define reference pulse shape with amplitude 1
- $\bar{n}_{norm} = \int |E_{ref,in}|^2 dt$
- Input pulse is then defined as: $E_{in} = A * E_{ref,in}$

$$\begin{aligned}\bar{n} &= \int |E_{in}|^2 dt \\ &= \int A^2 \cdot |E_{ref,in}|^2 dt \\ &= A^2 \cdot \bar{n}_{norm}\end{aligned}$$

$$\Rightarrow A = \sqrt{\frac{\bar{n}}{\bar{n}_{norm}}}$$

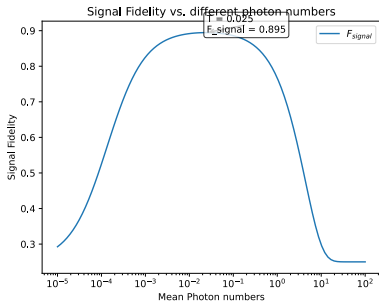
Photon Number Scan

- Mean photon number scanned logarithmically: $\langle n \rangle \in [10^{-5}, 10^2]$
- Covers:
 - Deep single-photon regime ($P_{sig} = \frac{\eta \bar{n}}{\eta \bar{n} + P_{dark}}$)
 - Intermediate few-photon regime
 - Onset of multiphoton effects (Dürr 2021)
- Final Fidelity:

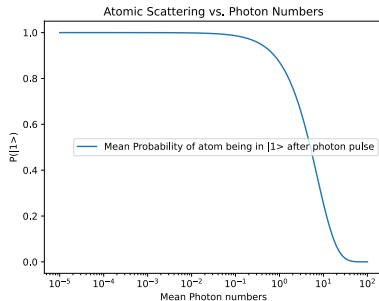
$$F = F_{\gg 1} + ([P_{sig} \cdot F_1 + (1 - P_{sig}) \cdot F_{\ll 1} - F_{\gg 1}]) \cdot e^{-(1-\eta)\bar{n}}$$

- $\bar{n} \ll 1 : e^{-(1-\eta)\bar{n}} \approx 1 \rightarrow F = P_{sig} \cdot F_1 + (1 - P_{sig}) \cdot F_{\ll 1} \approx F_{\ll 1}$
- $\bar{n} \gg 1 : e^{-(1-\eta)\bar{n}} \approx 0 \rightarrow F \approx F_{\gg 1}$

Results: Fidelity and Atomic Scattering vs Photon Number

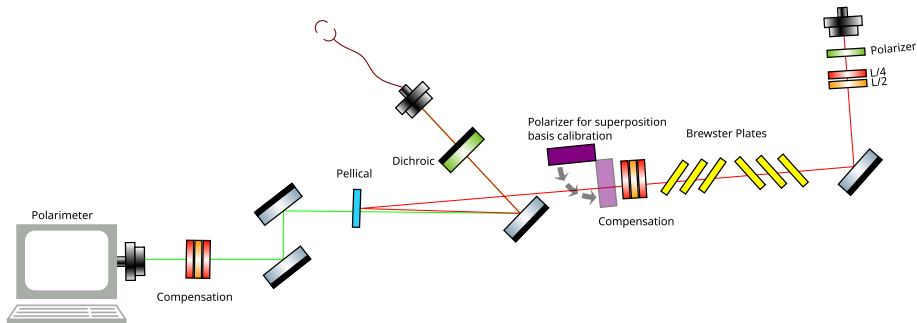


Fidelity vs $\langle n \rangle$



Atomic scattering vs $\langle n \rangle$

New Optical Table setup



Reflection Resonant Light

Reflection_Resonant_Light

